

Mr. Virgil: Learning Multi-robot Visual-range Relative Localization

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<https://github.com/HiOnes/Mr-Virgil>

Background

Visual-Range Localization:

- UWB-vision fusion localization has achieved extensive applications in the domain of multi-agent relative localization.

Challenge

Data Association

UWB Range

UAV Prior

Visual

Detection

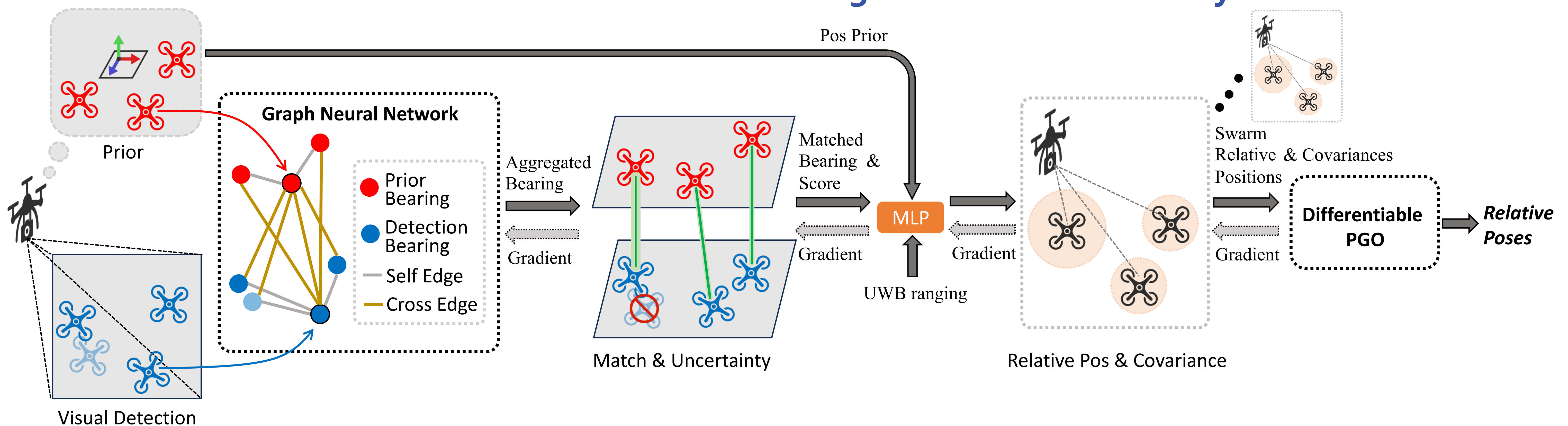
Existing Methods

- ID-encoded** (complex hardware time synchronization)
- Detection & tracking-based** (hand-crafted rule-based post-processing)

Overconfident yet erroneous matches may bring about irreversible damage to the localization system.

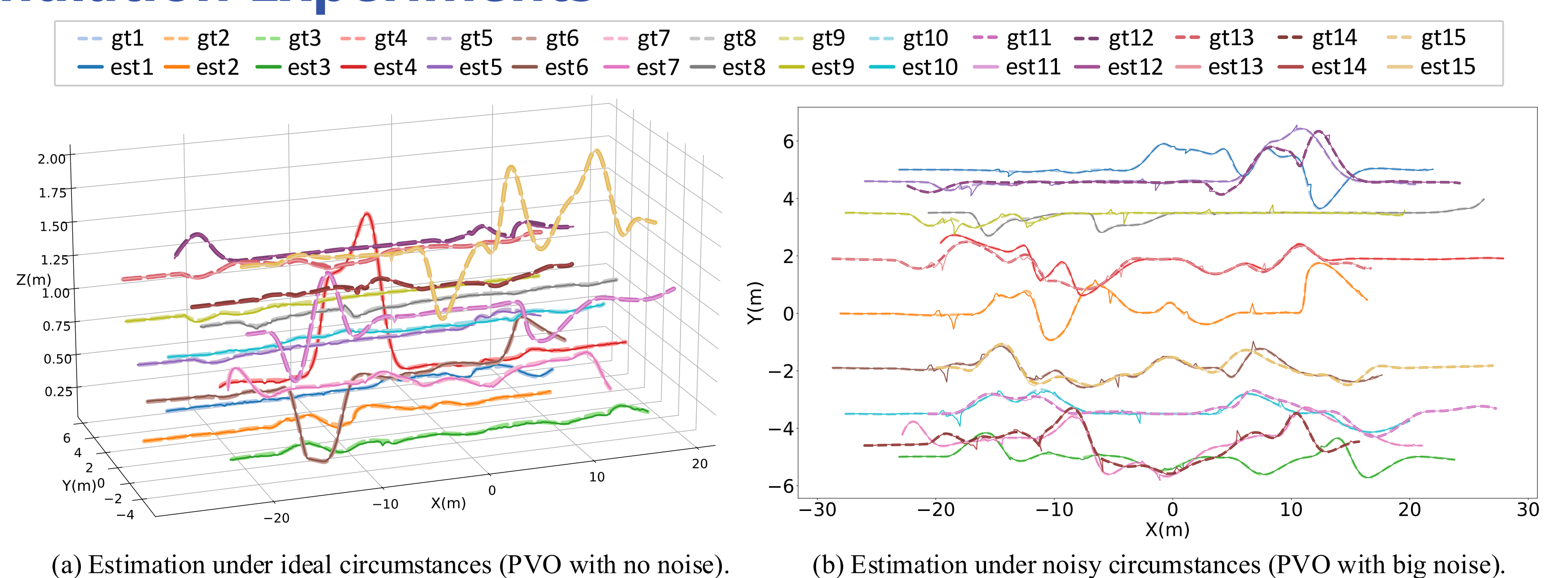
Frameworks

End-to-end multi-robot visual-range relative localization system

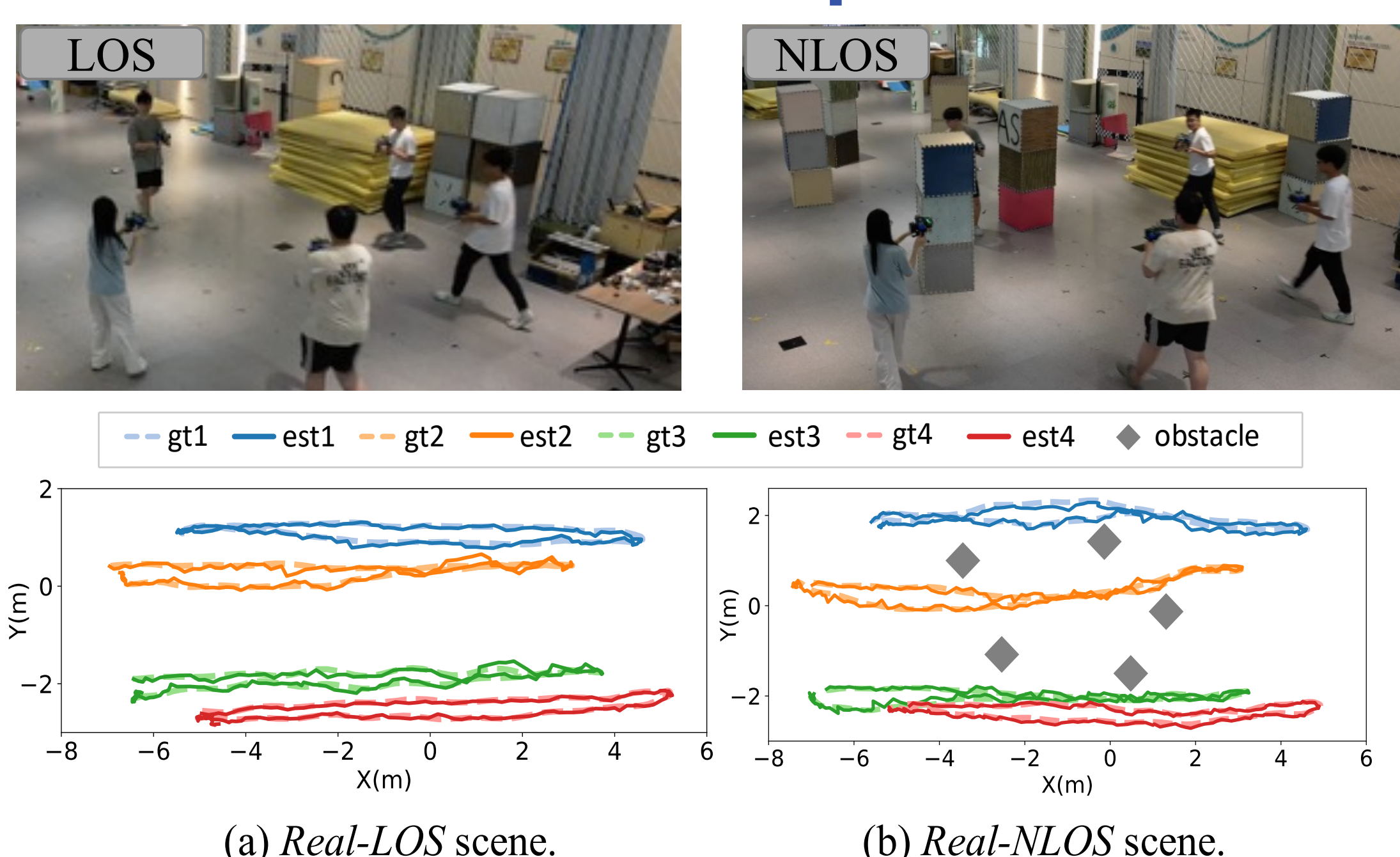


Experiments

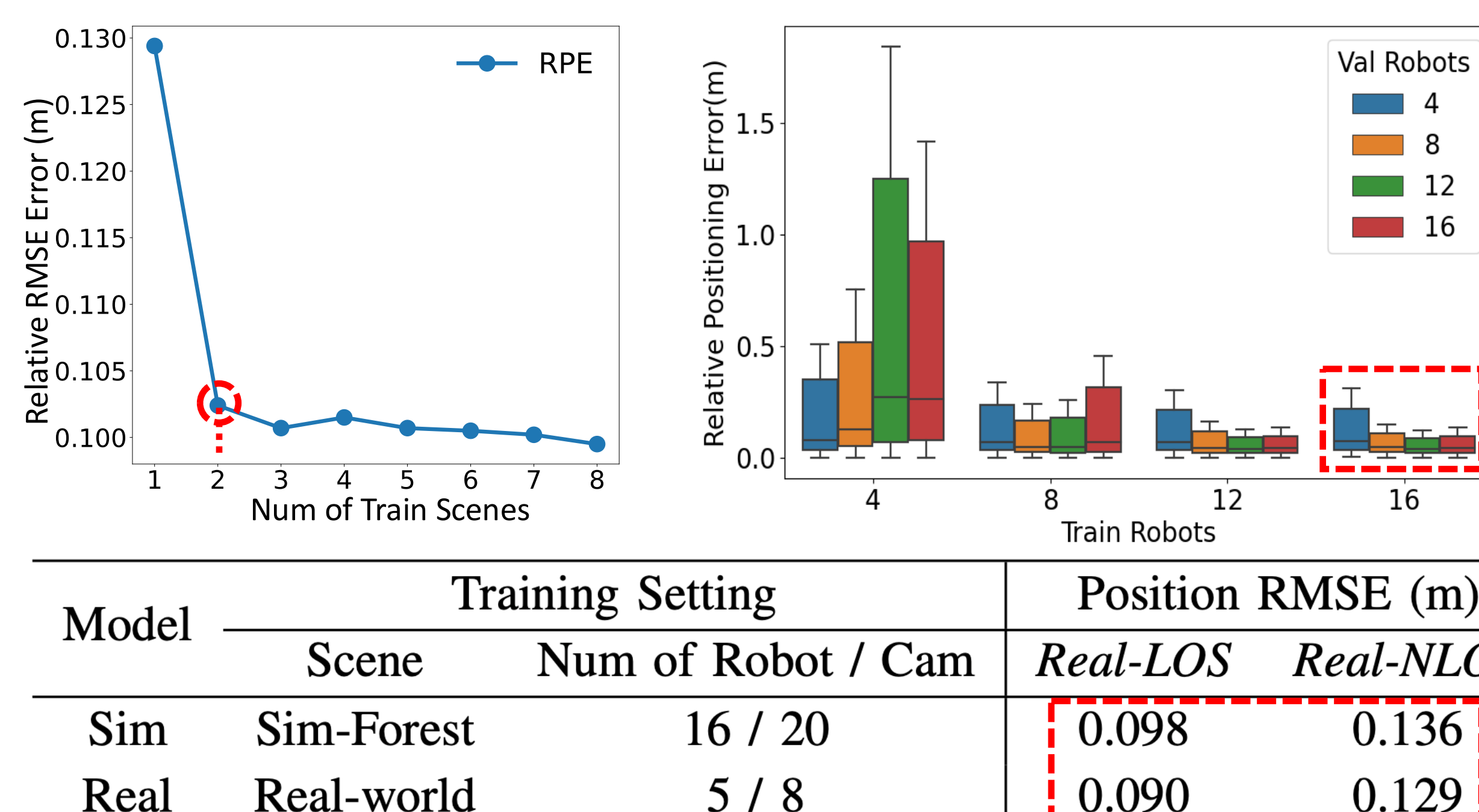
Simulation Experiments



Real-world Experiments



Network Generalization



- Training scenes**
 - Reaches comparable performance after being trained on only 2 scenes
- Robot Numbers**
 - Networks trained with a larger number of robots achieve higher accuracy
- Sim2Real**
 - The accuracy gap between the simulation model and the real model is less than 1cm